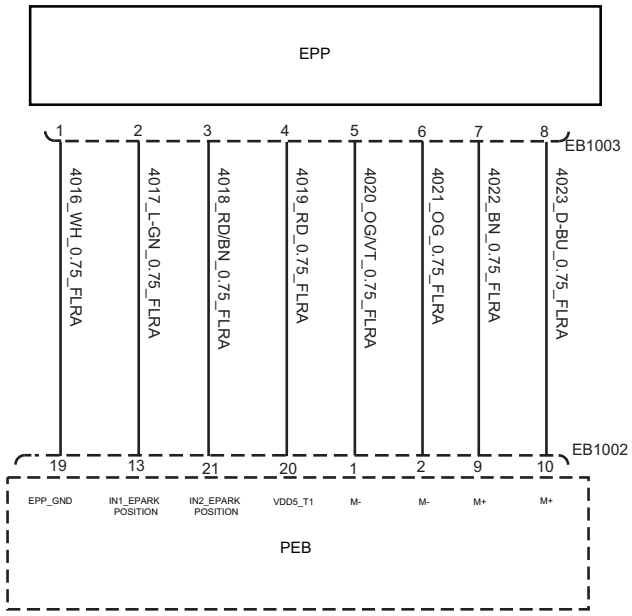
P193A-11、P193F-13、P193B-12、P193C-13、P193D-11、P193E-12、U0140-87、P1940-00、P1941-00

Fault diagnosis code
P1939-00：EPP1 way position sensor short circuit to ground
P193A-11：EPP1 way position sensor to power supply
P193B-12：EPP1 way position sensor open
P193C-13：EPP2 way position sensor short circuit to ground
P193D-11：EPP2 way position sensor to power supply
P193E-12：EPP2 way position sensor open
P193F-13：EPP1 way position feedback out of range
P1940-00：EPP2 way position feedback out of range
P1941-00：EPP two-way position feedback inconsistent fault
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • EPP fault • The EPP controller circuit in the PEU is faulty
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P1939-00
EPP1 way position sensor signal exceeds the STG threshold: 25
P193A-11
EPP1 way position sensor signal exceeds the STB threshold: 4070
P193B-12
EPP1 way position sensor signal exceeds the Open fault threshold: 25
P193C-13
EPP2 way position sensor signal exceeds the STG threshold: 25
P193D-11
EPP2 way position sensor signal exceeds the STB threshold: 4070
P193E-12
EPP2 way position sensor signal exceeds the Open fault threshold: ： 25
P193F-13
EPP1 way position sensor signal is greater than upper threshold 4070 or is less than lower threshold 25
P1940-00

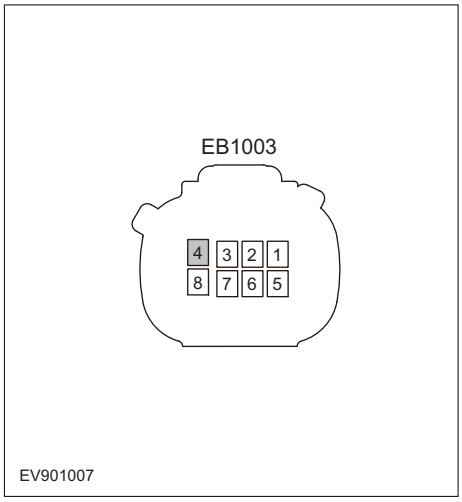
EPP2 way position sensor signal is greater than upper threshold 4070 or is less than lower threshold 25
P1941-00
The difference between the two senors is greater than the allowable error of calibration (calibration, default value 0.2)
To set the effect of a fault code condition
Parking system malfunction

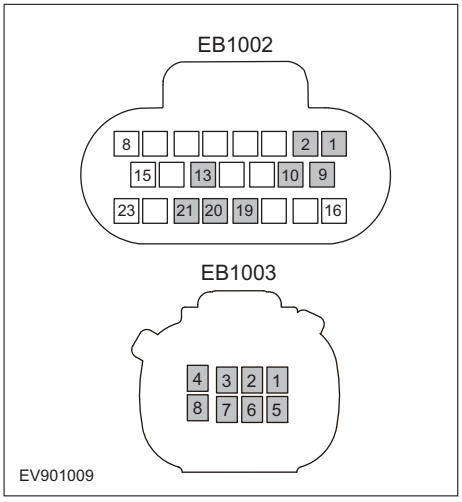
Circuit diagram



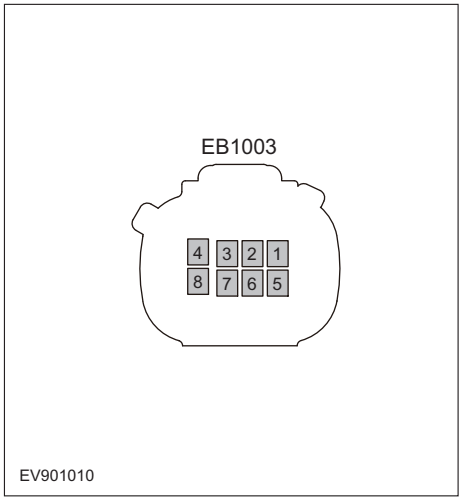


EV901006

Troubleshooting steps	
Step 1	General inspection.
	1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of PEU for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the PEU with diagnostic apparatus. 4 If there is any fault code except for P1939-00、P193A-11、P193B-12、P193C-13、P193D-11、P193E-12、P193F-13、 P1940-00、 P1941-00?
	YES → Refer to:DTC Summary list(PEU).
NO ↓	
Step 3	Check the power supply circuit of EPP circuit.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB1003 of EPP. 3 Turn the ignition switch to ON. 4 Measure the voltage between the harness terminal of EPP and ground. Measuring circuit: voltage between terminal 4 on plug EB1003 and ground. **Standard value:4.8~5.2V** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 4	Check the EPP circuit(open).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB1002 of PEU module. 3 Measure the resistance between the harness terminal of EPP and the PEU. Measuring circuit: resistance between terminal 1 on plug EB1003 and terminal 19 on plug EB1002. Measuring circuit: resistance between terminal 2 on plug EB1003 and terminal 13 on plug EB1002. Measuring circuit: resistance between terminal 3 on plug EB1003 and terminal 21 on plug EB1002. Measuring circuit: resistance between terminal 4 on plug EB1003 and terminal 20 on plug EB1002. Measuring circuit: resistance between terminal 5 on plug EB1003 and terminal 1 on plug EB1002. Measuring circuit: resistance between terminal 6 on plug EB1003 and terminal 2 on plug EB1002. Measuring circuit: resistance between terminal 7 on plug EB1003 and terminal 9 on plug EB1002. Measuring circuit: resistance between terminal 8 on plug EB1003 and terminal 10 on plug EB1002. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 5	Check the circuit of EPP (short).



1 Turn the ignition switch to OFF.

2 Measure the resistance between the harness terminal of EPP and ground.

Measuring circuit: resistance between terminal 1 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 2 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 3 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 4 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 5 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 6 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 7 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 8 on plug EB1002 and ground.

Standard value: ∞

3 Measure the voltage between the harness terminal of EPP and ground.

Measuring circuit: voltage between terminal 1 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 2 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 3 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 4 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 5 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 6 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 7 on plug EB1002 and ground.

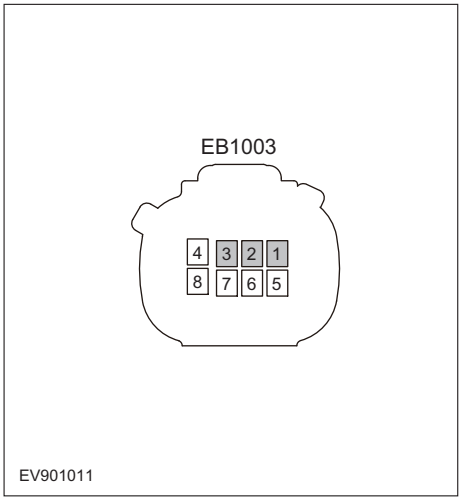
Measuring circuit: voltage between terminal 8 on plug EB1002 and ground.

Standard value: ≈ 0V

4 Check whether the result is normal or not?

NO → Repair or replace the faulted parts.

YES ↓

Step 6	Check the EPP (components).
	1 Measure the resistance of the EPP terminal directly. Measuring circuit: resistance between terminal 2 and terminal 1 on plug EB1003. Measuring circuit: the resistance between terminal 3 and 1 on plug EB1003. **Standard value:** 2 Check whether the result is normal or not?
	NO → Replace the EPP.
YES ↓	
Step 7	Test whether PEU is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1002 of PEU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. 4 Fly out of pins-install new pins as needed. 5 Reconnect the PEU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 6 Run the system and determine if the problem persists?
	YES → Change the PEU Module
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	